



Dayananda Sagar College of Engineering
An Autonomous Institute Affiliated to VTU, Belagavi
Department of Artificial Intelligence and Machine Learning

DATA SCIENCE

Course Code	22AI51	Credits	03
L:T:P:S	3:0:0:0	CIE Marks : SEE Marks	50:50
Course Type	PCC	Exam Hours	3

Course Objectives	
1	To provide the students with the basic knowledge of Data Science.
2	To enable students to acquire skills in data collection, data preprocessing & Data analysis effectively.
3	To make students have hands-on practice on visualization tools to extract meaningful insights and present data.
4	To foster awareness of the ethical issues in data handling, prepare students to be responsible data scientists.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Understand the foundational concepts of Data Science and various tools used in Data Science.	L1, L2
CO2	Apply data engineering techniques for the problem in hand.	L3
CO3	Analyze the data for the given application and extract valuable insights using data analysis techniques .	L4
CO4	Create visualization to effectively communicate data insights through storytelling while addressing ethical considerations in data handling.	L5

Mapping of Course outcomes to Program outcomes:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	-	-	-	-	3	-	-	-	-	-	-	2	-	2	-
CO2	3	-	-	-	-	2	-	-	-	-	-	-	-	2	-
CO3	-	3	-	-	-	-	-	-	-	-	-	-	-	2	-
CO4	-	-	-	3	3	2	-	-	-	2	-	-	-	2	-

Module	Topic	Hours	CO
1	Introduction: What Is Data Science? Big Data and Data Science Hype Getting Past the Hype Why Now? Datafication, The Current Landscape (with a Little History), Data Science Jobs, A Data Science Profile, Data Science Process Life Cycle, Applications of Data Science, Comparison of Data science with Data Analytics, Machine Learning, Business Intelligence, Modelling Process in Data Science. Data Acquisition: What is Data? Types of Data, Properties of Data, Types of data sets, Data Sources, Methods of Data Collection.	8	1,2,3,4
2	Data Visualization: Introduction to data visualization tools (e.g., Matplotlib, Seaborn, Tableau), Basic principles of creating effective visualizations- clarity, accuracy, relevance, simplicity, Importance and Examples of data storytelling. Exploratory Data Analysis and the Data Science Process: Basic tools (plots, graphs and summary statistics) of EDA, Philosophy of EDA, The Data Science Process, Case Study: RealDirect (online real estate firm). Data Preprocessing: Importance of Data Preprocessing, Key steps	8	1,2,3,4

	in data Preprocessing: Data Cleaning - Handling Missing Values, noise Data, Duplicate Data Outliers, Data Integration- Entity Identification Problem, Schema Integration, Detecting & resolving Data Value Concepts, Data Reduction – Dimensionality Reduction, Numerosity Reduction, Data Compression, Data Transformation- Smoothing, Aggregation, Discretization, Normalization.		
3	Data Mining: What is Data Mining? Types of Data Mining, Overview of the knowledge discovery process (KDD), Key goals of Data Mining: Prediction, Classification, Clustering, Association, and Anomaly Detection, Sequential Pattern discovery, Data mining Techniques, Data Mining Architecture, Applications of data Mining: - Market Basket Analysis: Discovering item purchase patterns in retail., Fraud Detection: Identifying suspicious transactions in banking and finance. Customer Segmentation: Dividing customers into distinct groups for targeted marketing. Healthcare: Identifying risk factors and predicting disease outbreaks. Web Scraping – Types of Web Scrappers, ETL Process, Web scrapping Tools (Examples like BeautifulSoup, Scrapy, Selenium, and Puppeteer.), Web Crawling. Advanced Scrapers: Bots that can navigate and extract data from complex, dynamic websites (AJAX, JavaScript). Data Warehousing: Data Warehouse Architecture- Top Down & Bottom-Up Approach, DBMS vs Data Warehouse, Components of Data Warehouse, ETL Process in Data warehouse.	8	1,2,3,4
4	Feature Generation and Feature Selection (Extracting Meaning from Data): Motivating application: user (customer) retention. Feature Generation (brainstorming, role of domain expertise, and place for imagination), Feature Selection algorithms. Manual Feature Selection, Correlation and Basic Statistics, Dimensionality Reduction (Concept Only), Filters; Wrappers; Building a User-Facing Data Product, Types of Features: - Categorical vs. Numerical: Identifying whether features are categories (e.g., customer location) or continuous numbers (e.g., order value). Derived Features: Creating new features by combining or transforming existing ones (e.g., creating a "loyalty score" based on purchase frequency). Time-Based Features: Creating features like "days since last purchase" or "average time between transactions" for applications like user retention.	8	1,2,3,4
5	Data Modelling: Data Modelling Types- Hierarchical model, Relational model, Network model, Object-oriented model, Entity-relationship model, Physical data model, Logical data model, Dimensional model, Steps in Data Modelling, - Requirement Gathering, Conceptual Modeling, Logical Modeling, Physical Modeling, Validation and Refinement. Tools in Data Modeling. - ERD Tool, Database Design Tools, Cloud-based Tools (AWS Glue and Azure Data factory) Data Science and Ethical Issues, Discussions on privacy, security, ethics, Next-generation data scientists. Sustainability and AI.	8	1,2,3,4

Text Books	
1	Doing Data Science Cathy O’Neil and Rachel Schutt Straight Talk from The Frontline.O’Reilly
2	Data Mining: Concepts and Techniques Jiawei Han, MichelineKamber and Jian Pei ThirdEdition

Reference Books	
1	Mining of Massive Datasets. V2.1 Jure Leskovek, AnandRajaraman and Jeffrey Ullman Cambridge University Press
2	Practical Statistics for Data Scientists Peter Bruce and Andrew Bruce O’reilly series



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ADVANCE MACHINE LEARNING

Course Code	21AI52	Credits	04
L:T:P:S	3:0:2:0	CIE Marks: SEE Marks	50:50
Course Type	IPCC	Exam Hours	3

Course Objectives

1	To introduce the fundamental techniques and principles of Neural Networks.
2	To study the different models in ANN and their applications.
3	To familiarize deep learning concepts with Convolutional Neural Network case studies.

Course Outcomes

		Blooms Taxonomy(highest)
1	Apply advanced forecasting, ensemble, recommendation, reinforcement learning techniques and optimization problems to solve complex real-world problems	L3
2	Analyze the performance and effectiveness of various models and algorithms	L4
3	Investigate and propose optimized solutions for decision-making processes	L4
4	Analyze multiple approaches to design innovative and ethical solutions	L4
5	Implement various analytical and predictive models	L3
6	Document the observations of implementing various models along their performances	L4

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2	3											1		
CO3		3												
CO4		3										1		
CO5									2					
CO6										3				

Module	Topic	Hours	Assessment	CO
1	Time Series Forecasting: Application of time series, Stationary of Data, Time Series Decomposing, Moving average, Exponential Smoothing, Holt- Winters Algorithm, ARIMA model and Components	8	CIE+SEE+Quiz	1,2,3,4,5
2	Ensemble Technique-1: Bagging, Boosting, Voting, GBM (Gradient Boosting Machines), Evaluating Ensemble classifier, Random Forest Algorithm, Gradient Boosting Algorithm, XG-Boost.	8	CIE+SEE+Quiz	1,2,3,4,5



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3	Recommendation System: Collaborative filtering techniques (user-based and item-based), content-based filtering using TF-IDF and matrix factorization methods, hybrid recommendation systems, evaluation metrics for recommendation quality.	8	CIE+SEE+Quiz	1,2,3,4,5
4	Formulation an Optimization Problem, categorizing maximization / minimization problems, adding constraints, simplex problem, business use cases.	8	CIE+SEE+Quiz	1,2,3,4,5
5	Introduction to Reinforcement Learning, Deep reinforcement learning architectures (e.g., Q-Networks), Ethical considerations in deploying RL models, and emerging trends in RL applications (RLHF – Reinforcement learning from Human Feedback)	8	CIE+SEE+Quiz	1,2,3,4,5

S. No	Programs	Hours	CO
1	Mastering Time Series Forecasting	2	5,6
2	Bagging Ensemble Learning techniques	2	5,6
3	Boosting Ensemble Learning techniques	2	5,6
4	Collaborative Filtering Systems	2	5,6
5	Content-based filtering methods	2	5,6
6	Hybrid Recommendation system	2	5,6
7	Maximization / Minimization problem formation	2	5,6
8	Adding constraints to the optimization problem	2	5,6
9	Reinforcement Learning Architectures	2	5,6
10	Ethical and Emerging Trends in RL	2	5,6

Text Books	
1	“Machine Learning: A Probabilistic Perspective” by Kevin P. Murphy
2	“Reinforcement Learning: An Introduction” by Richard S. Sutton and Andrew G. Barto
3	“Deep Reinforcement Learning Hands-On” by Maxim Lapan
4	Data Science from Scratch, First Principles with Python by Joel Grus, Deep Learning Ian Goodfellow, Yoshua Bengio, Aaron Courvill.
5	Deep Learning Ian Goodfellow, Yoshua Bengio, Aaron Courvill
6	Forecasting, principles, and practice, Rob J Hyndman

Reference Books	
1	"Information Theory, Inference, and Learning Algorithms" by David J.C. MacKay



NATURAL LANGUAGE PROCESSING

Course Code	22AI53	Credits	4
L:T:P:S	3:0:2:0	CIE Marks : SEE Marks	50:50
Course Type	IPCC	Exam Hours	03

Course Objectives

1	Understand the importance of natural language processing (NLP) and its applications in text analysis and generation.
2	Develop practical skills in conducting NLP experiments for tasks such as information retrieval and language generation.
3	Gain proficiency in implementing NLP algorithms for building language generation systems and machine translation models.
4	Learn to apply information retrieval techniques and lexical resources effectively in NLP tasks.
5	Explore advanced NLP topics and apply them in real-world scenarios to solve language processing challenges.

Course Outcomes	Blooms Taxonomy(highest)
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CO1	Apply fundamental NLP techniques to preprocess and analyze textual data, focusing on word-level and syntactic analysis.	L3
CO2	Analyze semantic structures and discourse for natural language understanding tasks.	L4
CO3	Analyze machine translation approaches, emphasizing Indian language characteristics.	L4
CO4	Create advanced NLP applications like Transformer-based chatbots and dialogue systems, incorporating design principles and ethical considerations.	L5
CO5	Apply NLP algorithms and tools, including NER and morphological analysis, to extract and generate structured information from unstructured data.	L3
CO6	Analyze diverse textual data using advanced NLP techniques to extract meaningful insights.	L4

CO-PO-PSO Mapping	
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Module	Topic	Hours	Assesment	CO
1	OVERVIEW AND LANGUAGE MODELING Overview: Origins and challenges of NLP, Language and Grammar, Processing Indian Languages, NLP Applications, Information Retrieval. Language Modeling: Various Grammar - based Language Models, Statistical Language Model, N- Grams Model, Minimum Edit Distance.	8	CIE+SEE+Quiz	1
2	WORD LEVEL AND SYNTACTIC ANALYSIS Word Level Analysis: Regular Expressions, Finite-State Automata, Morphological Parsing, Spelling Error Detection and correction, Words and Word Classes, Part-of-Speech Tagging. Syntactic Analysis: Context-free Grammar, Constituency, Parsing, Probabilistic Parsing.	8	CIE+SEE+Quiz	1,2
3	SEMANTIC ANALYSIS AND DISCOURSE PROCESSING Semantic Analysis: Meaning Representation, Lexical Semantics, Ambiguity, Word Sense Disambiguation. Discourse Processing: Cohesion, Reference Resolution, Discourse Coherence and Structure.	8	CIE+SEE+Quiz	3
4	NATURAL LANGUAGE GENERATION AND MACHINE TRANSLATION Natural Language Generation: Architecture of NLG Systems, Generation Tasks and Representations, Application of NLG, Machine Translation: Problems in Machine Translation, Characteristics of Indian Languages, Machine Translation Approaches.	8	CIE+SEE+Quiz	3,4
5	TRANSFORMERS AND LARGE LANGUAGE MODELS, CHATBOTS & DIALOGUE SYSTEMS The Transformer: A Self-Attention Network, The Language Modeling Head, Large Language Models with Transformers, Large Language Models: Generation by Sampling, Large Language Models: Training Transformers, Potential Harms from Language Models Chatbots & Dialogue Systems Properties of Human Conversation, Frame-Based Dialogue Systems, Dialogue Acts and Dialogue State, Chatbots, Dialogue System Design	8	CIE+SEE+Quiz	4

Text Books

1	Natural Language Processing and Information Retrieval, Tanveer Siddiqui, U.S. Tiwary, 1st Edition Oxford University press, 2019.
2	Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition, Daniel Jurafsky and James H Martin, 2nd Edition, Prentice Hall, 2019

Reference Books

1	Natural Language Understanding, James Allen, 2nd Edition, Benjamin /Cummings Publishing company, 2017
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Sl. No.	Topics	Course Outcome
1	Study of Python and basic commands to access text data. (from notepad, pdf, word documents, online)	CO5
2	A) Perform Pre-processing (Tokenization, stop word removal, unwanted character removal using Regex, stemming and lemmatization) of Text.	CO5
	B) Perform text pre - processing on a given corpus without using any pre - defined NLP packages.	CO5
3	A) Implement N-Gram model (Unigram, bigram, trigram extraction).	CO5
	B) Implement N -Gram model in python without using any predefined NLP packages. Note: use corpus of your own choice.	CO5
4	A) Implement Part-of-Speech (POS) Tagging.	CO6
	B) Write a Python program to perform Part-of-Speech (POS) tagging on a given text corpus without using any predefined NLP packages.	CO6
5	A) Implement chunking to extract Noun Phrases.	CO6
	B) Implement chunking to extract Noun and Verb phrases without using any pre defined NLP packages	CO6
6	Sentence completion with words or phrases using random prompts.	CO6
7	A) Using Machine Learning sentiment classification	CO5
	B) Implement machine learning sentiment classification without using any pre defined NLP packages:	CO5
8	A) Text summarization (Extractive and Abstractive)	CO5
	B) Implement Python programs for extractive and abstractive summarization without using predefined NLP libraries.	CO5
9	A) Perform Name Entity Recognition (NER) on given text.	CO6
	B) Perform NER without using any pre defined NLP packages	CO6
10	A) Perform Morphological Analysis.	CO6
	B) Perform Morphological analysis without using any pre defined NLP packages	CO6



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FULL STACK DEVELOPMENT LAB

Course Code	22AIL54	Credits	01
L:T:P:S	0:0:2:0	CIE Marks: SEE Marks	50:50
Course Type	PCCL	Exam Hours	3

Course Objectives	
1	MongoDB: Learn the basics of Mongo DB, a NoSQL database, including how to install & configure & interact with it using the MongoDB shell & Compass GUI
2	Express.js: Gain an understanding of Express.js, a web application framework for Node.js, focusing on routing, middleware & handling HTTP requests & responses
3	React.js: Dive into React.js, a Javascript library for building user interfaces, learning about components, state management, props, hooks & routing
4	Node.js: Explore Node.js a Javascript runtime environment, understanding its event-driven architecture, asynchronous programming modules & package management using npm or yarn

Course Outcomes		Blooms Taxonomy(highest)
CO1	Applying JavaScript fundamentals to solve real-world problems	L3
CO2	Apply functional React applications	L3
CO3	Develop Full-Stack Applications with ExpressJS, NodeJS, and MongoDB:	L3
CO4	Developing complete web applications using the MERN Stack	L3

LAB	Content	Hour
1	Introduction to JavaScript (Overview, Syntax, Placement) and Variables	2
2	JavaScript Conditional Statements and JavaScript Loop	2
3	JavaScript, CSS in HTML	2
4	Jquery Selectors	2
5	React functional component	2
6	React class component	2
7	React Events and Forms	2
8	React Routing	2



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9	JSON API in React	2
10	NoSQL Databases connection with react	2

Text Books	
1	Learning React, 2nd Edition by Alex Banks, Eve Porcello, O'Reilly Media, Inc.
2	Full-Stack React Projects - Second Edition, by Shama Hoque Packt
3	MongoDB: The Definitive Guide - Powerful and Scalable Data Storage, Third Edition by Shannon Bradshaw (Author), Eoin Brazil (Author), Kristina Chodorow (Author), O'Reilly Media, Inc



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COMPUTER NETWORKS AND COMMUNICATION

Course Code	21AI552	Credits	03
L:T:P:S	3:0:0:0	CIE Marks: SEE Marks	50:50
Course Type	PEC	Exam Hours	3

Course Objectives	
1	Acquire Profound Knowledge in Routing, Internetworking, and Congestion Control
2	Analyze and Compare Diverse Network Architectures and Topologies
3	Analyze and Compare Network Models, Emphasizing TCP/IP Protocol Suite and OSI Model
4	Analyze Packet Routing and Conduct Case Studies

Course Outcomes		Blooms Taxonomy(highest)
CO1	Analyze and gain the knowledge of routing, internetworking, congestion control and quality of service.	L4
CO2	Analyze and compare various network models, focusing on the TCP/IP protocol suite and OSI Model.	L4
CO3	Analyze Packet Routing Protocol and case studies.	L4
CO4	Analyze network, transport and application layer protocols and services	L4

CO-PO-PSO Mapping															
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1		3													
CO2		3										1	1		
CO3		3													
CO4		3											1		

Module	Topic	Hours	Assessment	CO
1	Introduction: Networks, Network types. Network Models: TCP / IP protocol suite, Addressing, The OSI Model. Transmission Modes: Parallel Transmission and Serial Transmission. Link Layer: Data Link Control (DLC): DLC Services, Data Link Layer Protocols, High Level Data Link Control (HDLC), and Point-to-Point Protocol (PPP): Framing, Transition phases. Media Access Control (MAC): Random Access: CSMA/CD, CSMA/CA.	8	CIE+SEE+Quiz	CO3, CO2
2	Network layer design issues: Store and Forward packet Switching, Services Provided to the Transport Layer Implementation of Connectionless Service, Implementation of Connection-Oriented Service, Comparison of Virtual Circuit and Datagram Subnets;	8	CIE+SEE+Quiz	CO1, CO2



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	Routing algorithms: Shortest Path Routing, Flooding, Distance Vector Routing, Link state Routing, Hierarchical Routing Broadcast Routing, and Multicast Routing.			
3	Congestion Control Algorithms: General Principles of Congestion Control, Congestion Prevention Policies, and Congestion Control in Virtual- Circuit Subnets, Congestion Control in Datagram Subnets, Load Shedding, Jitter Control; Quality Of Service: Requirements, Techniques for Achieving Good Quality of Service Integrated Services Differentiated Services. Internetworking: How networks differ, How networks can be connected Connectionless Internetworking, Tunneling Internetwork Routing, Fragmentation.	8	CIE+SEE+Quiz	CO1, CO4
4	The Network Layer in the Internet: The IP Protocol, IP Addresses, Internet Control Protocols, OSPF- Interior Gateway Routing Protocol, BGP- Exterior Gateway Routing Protocol, IPv6. The Transport Service : Services provided to the Upper Layers . The Internet Transport Protocols: Introduction to UDP, RPC, RTCP, Introduction to TCP. The TCP Service Model.	8	CIE+SEE+Quiz	CO4
5	The TCP Protocol: TCP protocol, TCP Segment Header, TCP Connection Establishment, TCP Connection Release. TCP Transmission Policy, TCP Congestion Control, TCP Timer Management. Application Layer: World Wide web and HTTP, FTP, Electronic Mail, Telnet.	8	CIE+SEE+Quiz	CO4

Text Books	
1	Data Communications and Networking, Behrouz A Forouzan, 5th Edition, 2013, Tata McGraw-Hill, ISBN – 9781259064753.

Reference Books	
1	Computer Networks, Andrew S Tanenbaum, 5th Edition, 2014, Pearson Education; ISBN - 978-81- 7758-165-2.
2	Computer Networking, a Top-Down Approach, James Kurose and Keith Ross, 6th Edition, 2013, ISBN- 13: 978-0-13-285620-1.
3	Data and Computer Communications, William Stallings, 8th Edition, 2009, Pearson Education, ISBN- 13: 978-0131392052.



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SOFT COMPUTATING

Course Code	22AI553	Credits	03
L:T:P:S	3:0:0:0	CIE Marks: SEE Marks	50:50
Course Type	PEC	Exam Hours	3

Course Objectives

1	To Understand the Fundamentals and Applications of Soft Computing.
2	To Gain Proficiency in Neural Network Models and Architectures.
3	To Develop an understanding of Fuzzy Logic and Inference Techniques.
4	To Understand the history and basic concepts of genetic algorithms.

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Apply neural networks, bidirectional associative memories and adaptive resonance theory for solving different engineering problems	L2
CO2	Apply fuzzy logic and reasoning to handle uncertainty and solve various engineering problems	L2
CO3	Apply genetic algorithms to combinatorial optimization problems	L3
CO4	Apply concepts of fuzzy vs. crisp sets, membership functions, fuzzy set operations and properties	L3

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2	3											1		
CO3	3													
CO4	3											1		

Module	Topic	Hours	Assessment	CO
1	Introduction to Soft Computing: Soft computing vs. hard computing, evolution of soft computing, features and types of soft computing, applications of soft computing, basics of machine learning Basic concepts of Neural Networks, Model of Artificial Neuron, Neural Network Architectures, Characteristics of neural networks, Learning Methods, early neural network architectures, Application domains.	8	CIE+SEE+Quiz	CO1
2	Associative Memory Networks: Auto correlators, hetero correlators: Kosko's discrete Bi-direction associative memory (BAM), Exponential BAM, Application of Character Recognition.	8	CIE+SEE+Quiz	CO2



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3	Fuzzy Sets and Fuzzy Relations: Fuzzy versus Crisp, Crisp Sets, Fuzzy sets, Membership functions, fuzzy set operations, properties of Fuzzy sets, Crisp Relations, Fuzzy relations – Fuzzy Cartesian product, Operations of Fuzzy Relations.	8	CIE+SEE+Quiz	CO3
4	Fuzzy Logic and Inference: Crisp Logic, Predicate Logic, Fuzzy Logic, Fuzzy Quantifiers, Fuzzy Inference, Fuzzy knowledge and rule-based system, fuzzy decision making, Defuzzification, Application of fuzzy logic	8	CIE+SEE+Quiz	CO3
5	Genetic Algorithm: History of Genetic Algorithm, Basic concepts, Creation of offspring, working principles, encoding, fitness function, reproduction, Genetic modeling: Inheritance operator, crossover, inversion & deletion, mutation operator, Bitwise operator, Generational Cycle, Convergence of GA, Applications & advances in GA, Differences & similarities between GA & other traditional method, Hybrid systems, evolutionary computing, Genetic Algorithm based on Backpropagation networks- Implementation and comparison on performance of traditional algorithms with Genetic Algorithm	8	CIE+SEE+Quiz	CO3, CO4

Text Books	
1	S, Rajasekaran& G.A. VijayalakshmiPai, “Neural Networks, Fuzzy systems and evolutionary algorithms: Synthesis and Applications”, PHI Publication, 2 ndEd. 2017.
2	Timothy J. Ross, “Fuzzy Logic with Engineering Applications”, John Wiley and Sons, 3 rded, 2011.
3	S.N. Sivanandam& S.N. Deepa, “Principles of Soft Computing”, Wiley Publications, 3rded, 2018.

Reference Books	
1	Jang, Jyh-Shing Roger, Chuen-Tsai Sun, and Eiji Mizutani. "Neuro-fuzzy and soft computing computational approach to learning and machine intelligence" Pearson, 1997.
2	Kosko, B., Neural Networks and Fuzzy Systems: A Dynamical Systems Approach to Machine Intelligence, PHI Publication, 1994.
3	George J. Klir, Fuzzy Sets and Fuzzy Logic: Theory and Applications, Prentice Hall, 2015
4	Rich E and Knight K, Artificial Intelligence, McGraw Hill Education; 3 rded, 2017.
5	S. Haykin, “Neural Networks and Learning Machines”, Pearson Education Inc., 3 rd Ed 2008.



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Theory of Computation

Course Code	22AI554	Credits	03
L:T:P:S	3:0:0:0	CIE Marks: SEE Marks	50:50
Course Type	PEC	Exam Hours	3

Course Objectives

1	To give an overview of the theoretical foundations of computer science from the perspective of formal languages
2	To illustrate finite state machines to solve problems in computing
3	To explain the hierarchy of problems arising in the computer sciences.
4	To familiarize Regular grammars, context free grammar.

Course Outcomes

		Blooms Taxonomy(highest)
CO1	To use basic concepts of formal languages of finite automata techniques	L2
CO2	To Design Finite Automata's for different Regular Expressions and Languages	L2
CO3	To Construct context free grammar for various languages	L3
CO4	To solve various problems of applying normal form techniques, push down automata and Turing Machines	L3
CO5	To participate in GATE, PGECET and other competitive examinations	L4

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1														
CO2	3											1		
CO3	3	3												
CO4		3										1		
CO5									2			2		

Module	Topic	Hours	Assessment	CO
1	FINITE AUTOMATA (FA): Introduction, Deterministic Finite Automata (DFA) -Formal definition, simpler notations (state transition diagram, transition table), language of a DFA. Nondeterministic Finite Automata (NFA)- Definition of NFA, language of an NFA, Equivalence of Deterministic and Nondeterministic Finite Automata, Applications of Finite Automata, Finite Automata with Epsilon Transitions, Eliminating Epsilon transitions, Minimization of Deterministic Finite Automata, Finite automata with output (Moore and Mealy machines) and Inter conversion.	8	CIE+SEE+Quiz	CO1



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2	REGULAR EXPRESSIONS (RE): Introduction, Identities of Regular Expressions, Finite Automata and Regular Expressions- Converting from DFA's to Regular Expressions, Converting Regular Expressions to Automata, applications of Regular Expressions. REGULAR GRAMMARS: Definition, regular grammars and FA, FA for regular grammar, Regular grammar for FA. Proving languages to be non-regular -Pumping lemma, applications, Closure properties of regular languages.	8	CIE+SEE+Quiz	CO2
3	CONTEXT FREE GRAMMER (CFG): Derivation Trees, Sentential Forms, Rightmost and Leftmost derivations of Strings. Ambiguity in CFG's, Minimization of CFG's, CNF, GNF, Pumping Lemma for CFL's, Enumeration of Properties of CFL (Proof's omitted).	8	CIE+SEE+Quiz	CO3
4	PUSHDOWN AUTOMATA: Definition, Model, Acceptance of CFL, Acceptance by Final State and Acceptance by Empty stack and its Equivalence, Equivalence of CFG and PDA. TURING MACHINES (TM): Formal definition and behaviour, Languages of a TM, TM as accepters,	8	CIE+SEE+Quiz	CO3
5	RECURSIVE AND RECURSIVELY ENUMERABLE LANGUAGES (REL): Properties of recursive and recursively enumerable languages, Universal Turing machine, The Halting problem, Undecidable problems about TMs. Context sensitive language and linear bounded automata (LBA), Chomsky hierarchy, Decidability, Post's correspondence problem (PCP), undecidability of PCP.	8	CIE+SEE+Quiz	CO3, CO4

Text Books	
1	John E. Hopcroft, Rajeev Motwani, Jeffrey D. Ullman (2007), Introduction to Automata Theory Languages and Computation, 3 rd edition, Pearson Education, India.

Reference Books	
1	K. L. P Mishra, N. Chandrashekar (2003), Theory of Computer Science-Automata Languages and Computation, 2 nd edition, Prentice Hall of India, India.



Research Methodology and IPR

Course Code	22RM56	Credits	03
L:T:P:S	2:2:0:0	CIE Marks: SEE Marks	50:50
Course Type	AEC	Exam Hours	3

Course Objectives	
1	To identify the importance of research.
2	To understand the research methodology, tools and techniques.
3	To interpret the results of Research and Propose recommendations.
4	To understand Intellectual Property rights.

Course Outcomes		Blooms Taxonomy(highest)
CO1	Outline the importance of conducting research, and Intellectual property rights	L2
CO2	Analyse different types of research problem and criteria for good research.	L2
CO3	Examine features and importance of research design including collection and sampling of data	L3
CO4	Develop Intellectual property rights documents and technical reports.	L3

CO-PO-PSO Mapping												
CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2					2	2	2		1
CO2	3	3	3	2				2	2	2		1
CO3	3	3	3	2	1			2	2	2		1
CO4	3	3	3	2	2			2	2	2		1

Note: AAT components must be given to satisfy attainment of PO 8, 9 and 10

Weightage values: 3 – Strongly matched, 2 – moderately matched, 1 – weakly matched, (---) not matched.

Module	Course Contents	Hours	COs
1	Meaning, Objectives and Characteristics of research - Research methods Vs Methodology - Types of research -Descriptive Vs. Analytical, Applied Vs. Fundamental, Quantitative Vs. Qualitative, Conceptual Vs. Empirical - Research process - Criteria of good research – Developing a research plan, Case Study and Practice Session.	8	CO1, CO2, CO3
2	Defining the research problem- Selecting the problem- Necessity of defining the problem - Techniques involved in defining the problem-Importance of literature review in defining a problem-Survey of	8	CO2, CO3



	literature-Primary and secondary sources-Reviews, treatise, monographs - patents-web and source- searching the web-Identifying gap areas from literature review- Development of working hypothesis, Case Study and Practice Session.		
3	Research design and methods –Research design–Basic Principles-Need of research design– Features of good design, Development of Models-Developing a research plan-exploration, Description, Diagnosis, and Experimentation – Determining experimental and sample designs, Sampling design -Steps in sampling design-Characteristics of good sample design-Types of sample designs- Measurement and scaling techniques- Methods of data collection -Collection of primary data-Data collection instruments Testing of hypotheses-Basic Concepts-Procedure for hypotheses testing flow diagram for hypotheses testing. Case Study and Practice Session.	8	CO1, CO2, CO3
4	Interpretation and report writing - Techniques of interpretation-Structure and components of scientific reports- Different steps in the preparation – Layout, structure and language of the report - Illustrations and tables- Types of report-Technical reports and thesis, Case Study and Practice Session IPRs - Role of IP in the Economic and Cultural Development of the Society, IP Governance, IP as a Global Indicator of Innovation, Invention and Creativity –Intellectual Property-Importance and Protection of Intellectual Property Rights (IPRs), Case Study and Practice Session.	8	CO1, CO4
5	A brief summary of: Patents, Copyrights, Trademarks, Industrial Designs-Integrated Circuits – Geographical Indications — Application and procedures. World Intellectual Property Organization (WIPO): Introduction, Origin of WIPO, Salient Features of WIPO, Main Activities of WIPO, India and WIPO, Access to International IP Filing Systems, Protection of Traditional Knowledge (TK), Indian web portals for patents & technologies, Case Study and Practice Session.	8	CO1, CO4

Text Books:

1	"Research Methodology: Methods and Techniques" (4 th Edition, 2019) by Kothari, C.R., New Age International.
2	"Intellectual Property A Primer for Academia" (1 st Edition, 2021) by Prof. Rupinder Tewari Ms. Mamta Bhardwaj.
3	"Research Design: Qualitative, Quantitative, and Mixed Methods Approaches" (6th Edition, 2023) by John W. Creswell and J. David Creswell
4	"Research Methodology: A Step-by-Step Guide for Beginners" (4th Edition, 2023) by Ranjit Kumar
5	"Intellectual Property Rights in an Age of Electronics and Information" (2023) by John V. Martin

Reference Books:

1	"Statistics with R: A Beginner's Guide" (2023) by Robert N. Stinerock
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2	"Research Methods in Education" by Louis Cohen, Lawrence Manion, and Keith Morrison (9th Edition, 2023):
3	"Intellectual Property in the New Technological Age: 2023" by Peter S. Menell, Mark A. Lemley, and Robert P. Merges
4	"Research Handbook on Intellectual Property and Moral Rights" edited by Paul Torremans (2023)

On-Line Materials & Resources (Video experiments / You-tube Videos / Virtual labs / Demo experiments videos, etc...):

1	https://link.springer.com/book/10.1007/978-981-19-5441-2
2	https://onlinecourses.swayam2.ac.in/ntr24_ed08/preview
3	https://ayushnext.ayush.gov.in/index.php/detail/writeUps/national-level-workshop-on-research-methodology-and-ipr
4	https://onlinecourses.nptel.ac.in/noc23_ge36/preview



Dayananda Sagar College of Engineering
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ENVIRONMENTAL STUDIES

Course Code	22ES57	Credits	02
L:T:P:S	2:0:0:0	CIE Marks: SEE Marks	50:50
Course Type	HSMS	Exam Hours	2

Course Objectives

1	Develop sense of responsibility about the environment, natural resources, their conservation
2	Understand the concept of various components of environment, their pollution and their remediation methods.
3	To study global environmental issues.

Course Outcomes

		Blooms Taxonomy(highest)
CO1	Understand the components of environment.	L2
CO2	Identify the various sources of energy.	L2
CO3	Discuss the various types environmental pollution.	L2
CO4	Interpret the waste management.	L3
CO5	Analysis the various types of special waste management.	L4
CO6	Summarize the global Environment issues.	L3

CO-PO-PSO Mapping

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
CO1	3	-	-	-	-	1	1	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	1	1	-	-	-	-	-	-	-	-
CO3	3	-	-	-	-	1	1	-	-	-	-	-	-	-	-
CO4	3	-	-	-	-	1	1	-	-	-	-	-	-	-	-
CO5	3	-	-	-	-	1	1	-	-	-	-	-	-	-	-
CO6	3	-	-	-	-	1	1	2	-	-	-	-	-	-	-

Module	Contents	Hours	CO's
1	Introduction: Environment – Definition, components of environment Ecosystem: types & structure of ecosystem, SDG goals. Water resources: sources, quantity and quality, <i>water borne diseases</i>	04	CO1
2	Energy: Different types of energy, Conventional & Non-conventional sources of energy: solar energy, hydro electric energy, wind energy, nuclear energy, CNG, <i>biomass & biogas</i> , fossil fuels, hydrogen as an alternative energy.	04	CO2
3	Environmental Pollution -water pollution, air pollution, noise pollution, and soil pollution - definition, sources, effects and remedies.	06	CO3
4	Solid Waste Management: Types of solid waste, E-waste, hazardous waste & biomedical waste, construction and demolition waste –definition, sources, & its effects.	06	CO4 & CO5



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5	Global Environmental Issues: Global warming, CO ₂ reduction acid rain & ozone layer depletion, Greenhouse gases, Regulations-commissions and Acts on EIA	06	CO6
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Self-Study Component		
Module	Contents	CO's
1	Water borne diseases	CO1
2	biomass & biogas	CO2
3	waste to energy concept	CO3
4	Biomedical waste	CO4, CO5
5	CO ₂ reduction	CO6

Text Books

1	S.M. Prakash, "Environmental Studies", Elite Publishers Mangalore, 2007
2	Benny Joseph (2005), "Environmental Studies", Tata McGraw – Hill Publishing Company Limited.
3	R. J. Ranjit Daniels and Jagadish Krishnaswamy, (2009), "Environmental Studies",
4	Wiley India Private Ltd., New Delhi.
5	R Rajagopalan, "Environmental Studies – From Crisis to Cure", Oxford University Press, 2005.

References

1	P. Meenakshi, " Elements of Environmental Science and Engineering ", Prentice Hall of India Private Limited, New Delhi, 2006
2	Aloka Debi, " Environmental Science and Engineering ", Universities Press (India) Pvt. Ltd. 2012
3	G.Tyler Miller Jr., "Environmental Science – working with the Earth", Eleventh Edition, Thomson Brooks /Cole, 2006

Important Websites/ Links:

1	https://www.unep.org/
2	https://www.epa.gov/
3	https://www.nature.org/en-us/
4	https://www.wri.org/
5	https://www.footprintnetwork.org/